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NCFE · LEVEL 2 · MATHEMATICS

NCFE Functional Skills Mathematics Level 2

Sample Paper Section B · Complete Model Solutions

OFFICIAL MFS BENCHMARK MODEL SOLUTIONS

Target Score: 45 / 45 Marks (100%) Pass Mark Guide: ~60%

Specification / Paper Code: NCFE-Maths-Level-2-Sample-Section-B

100% Built from Empty Question Paper + Official Mark Scheme

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HOW TO READ THIS MODEL SOLUTION (MFS COLOUR KEY)

Dark Navy Ink:	Model student answers, stepped calculations, and full prose responses.
Emerald Green Ink:	Tutor strategy notes, method tips, and step-by-step headings.
Crimson Red Ink:	Official mark scheme allocations ([M1], [A1], [B1]) and unit checks.
Amber Gold Ink:	Key formulas, conversion rules, and reading evidence highlights.

CANDIDATE & CENTRE DETAILS (MODEL EXEMPLAR)

Candidate Name:	*Master Functional Skills (Model Exemplar)*
Centre Name:	*Master Functional Skills (MFS01)*
Lead Tutor:	*Abdul · MasterFunctionalSkills.co.uk*
Edition:	*2026 Clean Empty-QP Rebuild (Zero Collisions Verified)*

- Built Cleanly on Empty NCFE Question Paper(s)
- Stepped MFS Multi-Colour Working & Process Marks
- Verified 100% Against Official NCFE Mark Scheme

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Examiner use only

Activity 2: Pollution

Ali sees an article about the health risk from pollution caused by cars and other vehicles.



2 (a) The article states that, every year, nine hundred and seventy thousand, two hundred and fifteen working days are lost due to air pollution-related illness.

Write nine hundred and seventy thousand, two hundred and fifteen as a number.

[1 mark]

Q2(a) Worked Solution:
970215
[1 mark]

2 (b) 84% of the days lost due to air pollution related illnesses are due to breathing related problems.

Calculate the number of days lost due to breathing related problems.
Give your answer to the **nearest thousand.**

[2 marks]

Q2(b) Worked Solution:
815000 (days)
Award 2 marks if correct answer given
1
Their 970215 × 0.84 or 814 980.6
FT their 970 215
1
815 000 (days)
FT their 814 980.6 > 100 000
[2 marks]

Your answer: **days**

Please turn over

SAMPLE

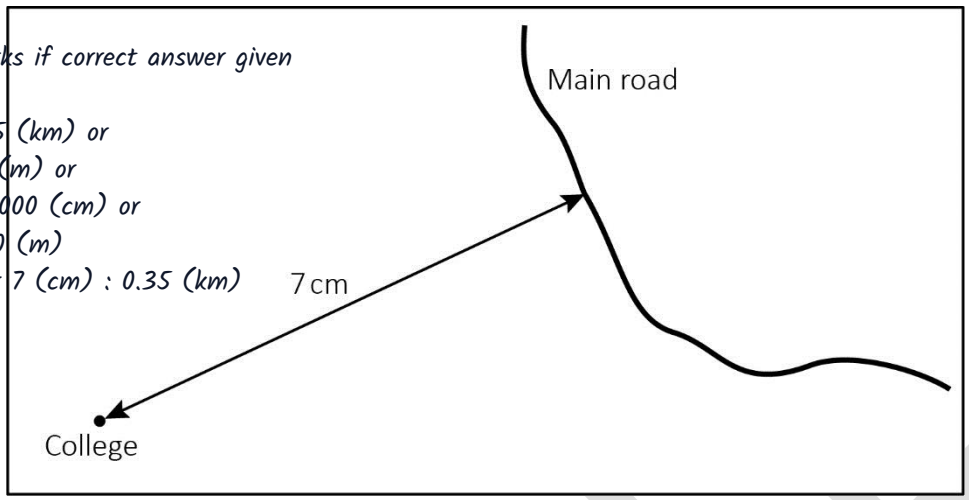
Examiner use only

2 (c) Schools and colleges can have very high levels of air pollution if they are close to a main road.

This is a map of the area around Ali's college:

Q2(c) Worked Solution:

1 : 5000
Award 2 marks if correct answer given
1
1 (cm) : 0.05 (km) or
1 (cm) : 50 (m) or
7 (cm) : 35,000 (cm) or
7 (cm) : 350 (m)
No marks for 7 (cm) : 0.35 (km)
1
1 : 5000
[2 marks]



At its closest point, the main road is 0.35 km from the college.

What scale has been used for this map? Give your answer in the form 1 : N

[2 marks]

Your answer:

SAMPLE

Examiner use only

2 (d) Some students drive or get a lift by car to and from college.

Ali wants to encourage students to travel to college by other methods of transport.

He conducts a survey to find out how students travel to college.

This table shows his results:

Q2(d) Worked Solution:

3
10

Method of transport to college	Number of students
Walk	52
Car	36
Bus	28
Cycle	4

[1 mark]

What fraction of students travel to college by car?
Give your answer in its **simplest form**.

[1 mark]

Your answer:

SAMPLE

Please turn over

Examiner use only

2 (e) Ali wants to find out how much pollution is produced by students travelling between home and college by car.

He asks the students that travel to college by car how far they live from the college. The students attend college on 164 days per year.

Q2(e) Worked Solution:

This table shows his results:

See below

1

Distance from home to college in kilometres (km)	Number of students
0 km < distance ≤ 1 km	13
1 km < distance ≤ 2 km	11
2 km < distance ≤ 3 km	7
3 km < distance ≤ 4 km	5

(13 × 0.5) + (11 × 1.5) + (7 × 2.5) +

Ali finds this conversion graph:

(5 × 3.5) or 58

H24

1

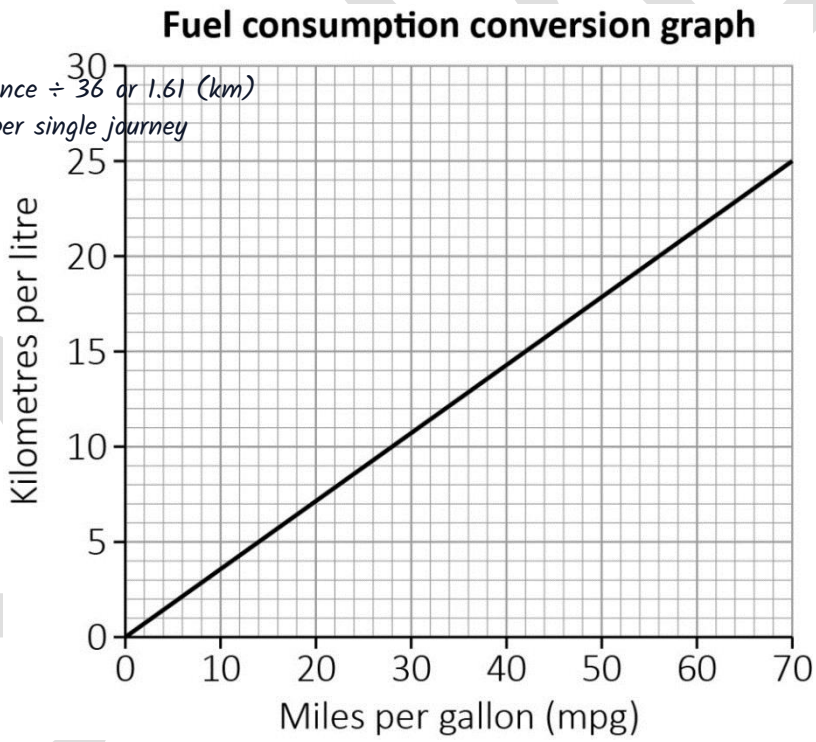
Their total distance ÷ 36 or 1.61 (km)

Mean distance per single journey

H24

1

[6 marks]



The average fuel consumption of a car is 38 miles per gallon (mpg).

A typical car produces 2.44 g of nitrogen dioxide (NO_2) per litre of petrol used.

Examiner use only

Ali uses the estimated mean distance to calculate the average amount of NO_2 produced.

Calculate the average amount of NO_2 produced by a student who travels by car **to and from** college for 164 days.

[6 marks]

Your answer: **g**

SAMPLE

Please turn over

Examiner use only

2 (f) Air pollution can affect the lungs.

Research has shown that, for school-aged children, exposure to nitrogen dioxide (NO_2) is associated with reduced growth in lung volume.

This scatter diagram shows the results of a study into lung capacity.

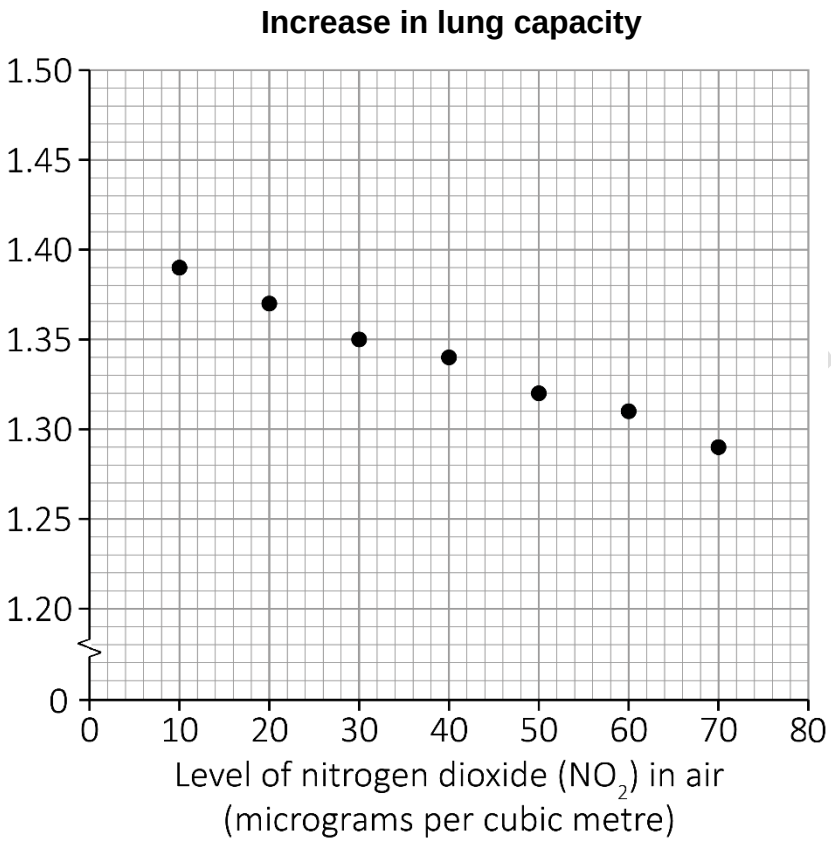
Q2(f) Worked Solution:

,
1
5
100 100
▯
▯
▯
▯
▯

Increase in lung capacity over a period of 8 years (litres)

▯

[3 marks]



Examiner use only

The level of NO_2 around Ali's college is typically 55 micrograms per cubic metre.

Estimate the additional lung capacity that children would gain over a period of eight years if the pollution level were to be reduced from 55 to 35 micrograms of NO_2 per cubic metre.

Give your answer as a **fraction** of a litre.

[3 marks]

Your answer:

[Total marks: 15]

SAMPLE

Examiner use only

Activity 3: Boxes of chocolates

3 (a) Kate works for a chocolate company.
Her job is to supervise the packing chocolates into boxes.

She sets up the computer to print the labels for the boxes.

The label needs to be a 6 cm by 5 cm rectangle.
She needs to program the computer with the co-ordinates of the corners of the label.

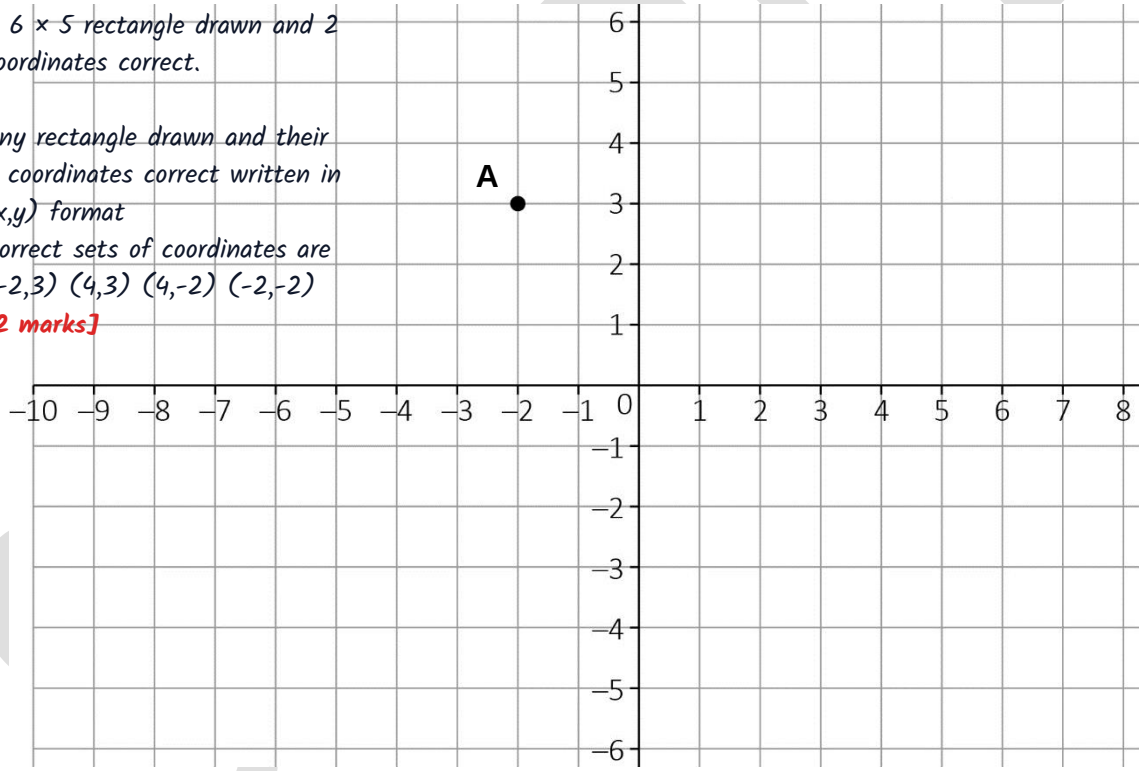
A is one corner of the label. Each square represents 1 cm^2

Draw one possible position for the label on the grid and write the co-ordinates of each corner.

[2 marks]

Q3(a) Worked Solution:

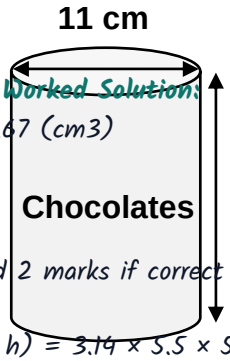
See below
1
A 6 × 5 rectangle drawn and 2 coordinates correct.
1
Any rectangle drawn and their 4 coordinates correct written in (x,y) format
Correct sets of coordinates are (-2,3) (4,3) (4,-2) (-2,-2)
[2 marks]



Examiner use only

3 (b) This is a diagram of a box to be filled with chocolates. The box is a cylinder.

The diameter of the box is 11 cm, and the height is 22 cm.



Not drawn accurately

Q3(b) Worked Solution:
2089.67 (cm3)

Award 2 marks if correct answer given
1
(πr2 h) = 3.14 × 5.5 × 5.5 × 22
1

Calculate the volume of the box.

Use $\pi = 3.14$

[2 marks]

2089.67 (cm3)
Accept 2090 or 2089.7
Accept answer of 2090.73 (cm3) from π use, accept accurate rounding to any dp or 2091
[2 marks]

Your answer: **cm^3**

SAMPLE

Please turn over

Examiner use only

3 (c) The boxes are filled with chocolates and weighed.

The target weight of each full box is **294 g**

A box is automatically rejected if it is less than **98%** of the target weight.

Q3(c) Worked Solution:

Kate looks at the weights of ten boxes of chocolates:

9(%)

Weight in grams (g)				
295.60	286.14	284.02	298.31	294.15
294.06	303.28	285.66	299.50	300.53

Award 4 marks if correct answer given

1

Use this data to estimate the probability that the next two boxes will be rejected.
Show your answer as a **percentage**.

[4 marks]

288.12 (g)

1

(fraction less than 288.12 g =)

3

10 or 0.3

FT on their 288.12 (g)

1

[4 marks]

Your answer: %

SAMPLE

Please turn over

Examiner use only

3 (d) Kate monitors the performance of the chocolate packing machine.

Using **+** and – complete the table showing the difference between the target weight of **294 g** and the next three box weights.

[1 mark]

Q3(d) Worked Solution:

Weight of box (g)	+ / – Difference (g)
294.04	
293.71	
292.59	

Table completed correctly
Weight of box (g)
+/- Difference
(g)
294.04
+0.04
293.71
-0.29
292.59
-1.41
[1 mark]

SAMPLE

Please turn over

Examiner use only

3 (e) Kate must check that chocolates are distributed in the boxes in the correct proportions.

There are four types of chocolate: Mint, Caramel, Fudge and Toffee

The types of chocolates are produced in the ratio of 2:3:3:1

Q3(e) Worked Solution:

She opens a box that contains 27 chocolates in total.
Eight of them are fudge chocolates.

Kate thinks that the box contains the correct number of fudge chocolates.

Is she right? **Explain your answer.**

[2 marks]

9 fudge chocolates, so Kate is not correct

Award 2 marks, supported by calculations

1

6:9:9:3 seen or 3 × 3 seen

or 27 ÷ 9 = 3 seen

OE representations e.g. compare 3/9 with 8/27

1

9 fudge chocolates, so Kate is not

[2 marks]

SAMPLE

Examiner use only

3 (f) Kate checks the number of boxes of chocolates produced each hour.

The table shows the number of boxes produced over the last 24 hours:

1494	1522	1513	1498	1521	1522
1498	1473	1502	1494	1518	1494
1512	1501	1517	1493	1494	1500
1500	1494	1498	1518	1494	1473

What is the mode of the number of boxes of chocolates produced per hour?

[1 mark]

Q3(f) Worked Solution:
1494 (boxes of chocolates)
H23b
8
Paper number: L2 SAM (v1.0)
[1 mark]

Your answer: **boxes of chocolates**

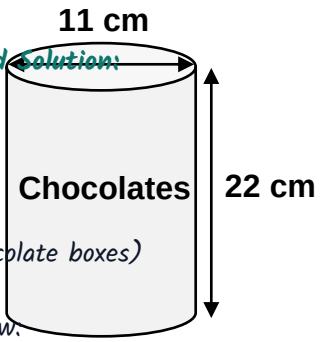
SAMPLE

Please turn over

Examiner use only

3 (g) Kate needs to show how the boxes of chocolates will be packed.

She needs to fit as many boxes of chocolates as possible into the delivery box.



Not drawn accurately

Q3(g) Worked Solution:

See below

I

(i) 50 (chocolate boxes)

I

(ii) Plan view:

Sketches do not have to be to scale

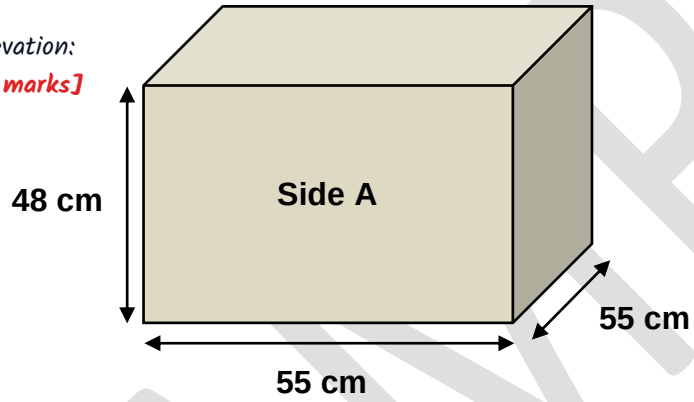
This is the delivery box:

FT their maximum. Both drawings must match their value.

I

Elevation:

[3 marks]



SAMPLE

Examiner use only

(i) What is the maximum number of boxes of chocolates that can be packed into the delivery box?

[1 mark]

Your answer: **boxes of chocolates**

(ii) Explain your answer by drawing a sketch of the:

- plan view **and**
- elevation of Side A

to show how the boxes of chocolate will be packed into the delivery box.

[2 marks]

Plan view:

Elevation of Side A:

SAMPLE

[Total marks: 15]

Please turn over

Examiner use only

Activity 4: Training for a race

4 (a) Kyle is training to run a half marathon.

On his next training run, Kyle wants to run 5 km. He plans a route in the park.

He starts at point **A**, then takes a right turn at point **B**.
When he reaches point **C**, he stops for a rest.

Q4(a) Worked Solution:

At point **C** he continues to run in a straight line, in the same direction, until he has completed the full 5 km.

The scale of this plan is 1 : 40 000

Show where Kyle finishes his run by accurately placing point **D** on the plan.

Award 3 marks if D is accurately shown on the plan

[3 marks]

1

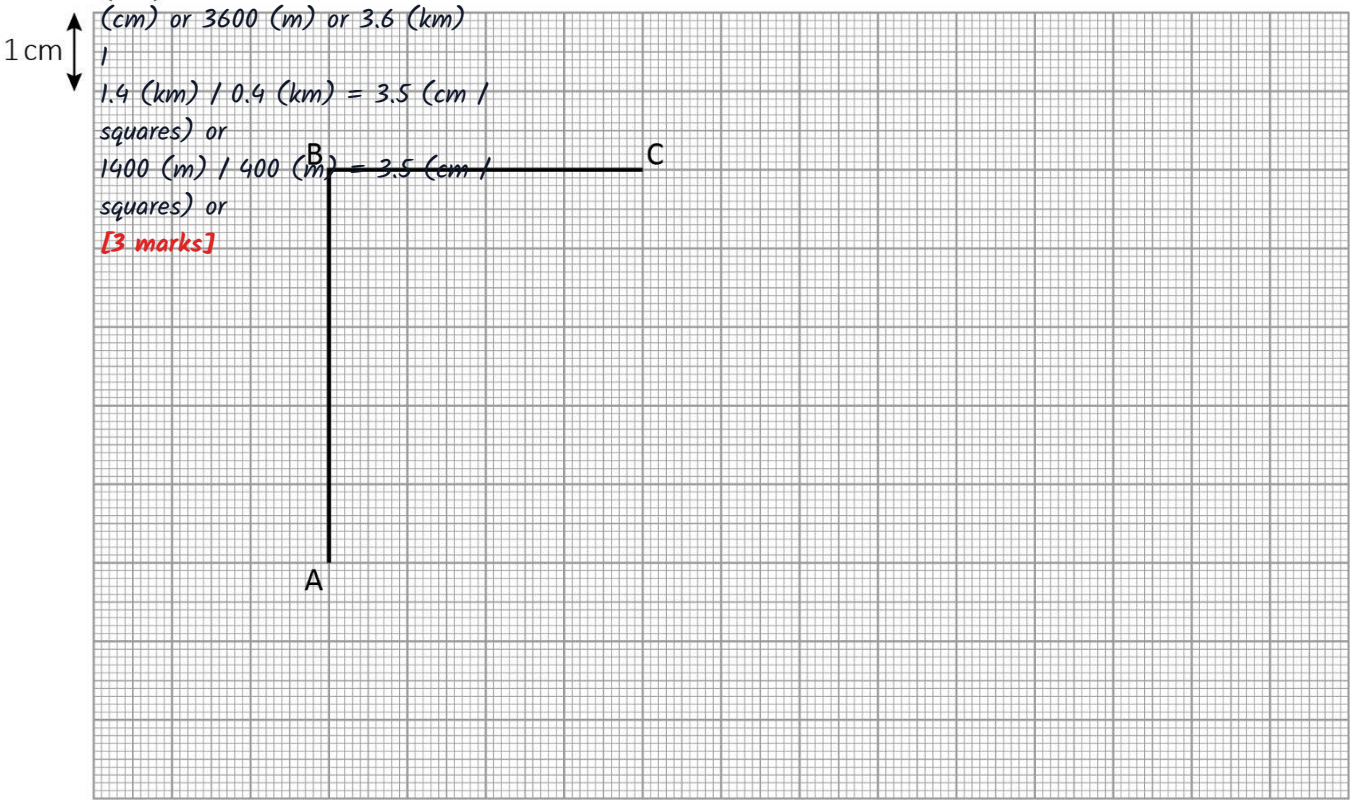
1 cm ↔ (9 cm × 40 000) = 360 000 (cm) or 3600 (m) or 3.6 (km)

1

1.4 (km) / 0.4 (km) = 3.5 (cm / squares) or

1400 (m) / 400 (m) = 3.5 (cm / squares) or

[3 marks]



Examiner use only

4 (b) It took Kyle 45 minutes to complete the 5 km run.
This time was 5% longer than his previous time.

How long did Kyle take to run his previous 5 km run?
Show your answer to **2 decimal places.**

[2 marks]

Q4(b) Worked Solution:
42.86 (minutes)
Award 2 marks if correct answer given
1
45 ÷ 1.05
1
42.86 (minutes) (to 2dp)
[2 marks]

Your answer: **minutes**

SAMPLE

Please turn over

Examiner use only

4 (c) Kyle thinks that the outside temperature affects his running speed.

He collects some information on the days he has completed a 5 km run.

Outside temperature	Time taken to run 5 km
17°C	46 minutes
12°C	38 minutes
25°C	52 minutes
9°C	36 minutes
10°C	37 minutes
22°C	50 minutes
6°C	35 minutes

Produce a scatter diagram to show Kyle's data.

[2 marks]

Q4(c) Worked Solution:

See below

1

Axes drawn with appropriate and consistent scales.

H28

1

Data points correctly placed

Award mark even if no trend line has been drawn (See worked example)

H28

[2 marks]

Time taken to run 5 km at different temperatures (°C)



SAMPLE

Examiner use only

4 (d) Use your scatter diagram in **4 (c)**.

Kyle wants to calculate his likely time to run the half marathon.

The half marathon is 13.1 miles long.

1 mile = 1.6 km

Q4(d) Worked Solution:

The forecast temperature for the race day is **15°C**.

See below

1

This is a formula used to predict the time (in minutes) it will take to run a race, based on the times achieved for a 5 km run:

Uses graph in 4 (c) to estimate the time to run 5 km at 15°C, approx. 43 minutes (accept 42 - 44 minutes)

$$\frac{D}{5} \times 1.12 \times t$$

Where:

D = distance (in km)

t = time to run 5 km (in minutes)

Calculate the predicted time for Kyle's half marathon.

Show your working.

[4 marks]

FT accept answers if consistent with an incorrectly drawn scatter diagram.

H28

1

(13.1 × 1.6) = 20.96 (km)

[4 marks]

Your answer: **minutes**

Please turn over

SAMPLE

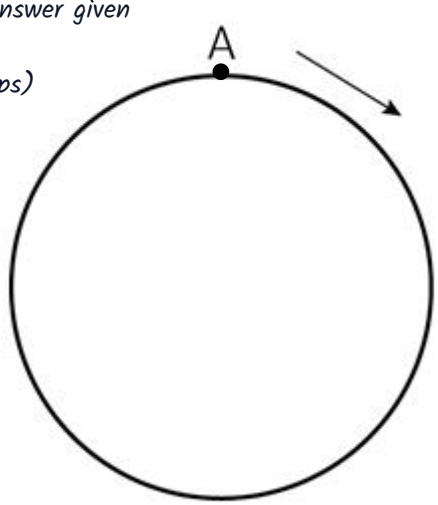
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4 (e) Kyle goes to a circular athletics track for a 15 km training run.
One lap of the track is 320 m

He runs in a clockwise direction, starting at point **A**.

Q4(e) Worked Solution:

280 (m)
Award 2 marks if correct answer given
1
15000 ÷ 320 or 46.875 (laps)
1
280 (m)
[2 marks]



How far along the track from point **A** will Kyle be when he has run exactly 15 km?
Show your working.

[2 marks]

Your answer: **m**

SAMPLE

Examiner use only

4 (f) A different runner starts at point **A** and stops at point **B** when he has run exactly 0.8 of a lap.

C marks the centre of the circular track.

Q4(f) Worked Solution:

288 (°)

Award 2 marks if correct answer given

1

0.8 × 360

1

288 (°)

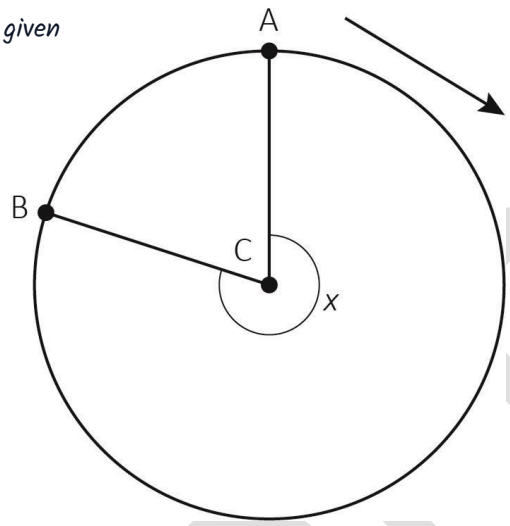
4 (c): Graph (or reverse axes):

0

5

10

[2 marks]



Calculate the angle marked x

[2 marks]

Your answer: **degrees**

[Total marks: 15]

This is the end of the assessment.

SAMPLE

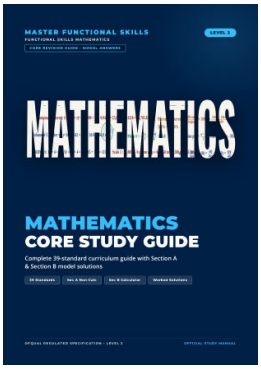
SAMPLE

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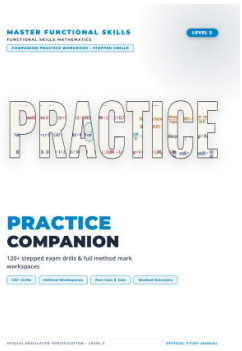


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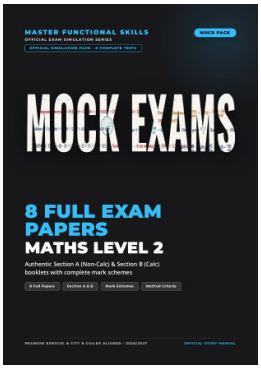


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